

$$5. A = \begin{bmatrix} 1 & -2 & -4 & -3 \\ 2 & -7 & -7 & -6 \\ -1 & 2 & 6 & 4 \\ -4 & -1 & 9 & 8 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 7 \\ 0 \\ 3 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ -4 & 3 & -5 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 & -4 & -3 \\ 0 & -3 & 1 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Solution: $L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ -4 & 3 & -5 & 1 \end{bmatrix} \quad U = \begin{bmatrix} 1 & -2 & -4 & -3 \\ 0 & -3 & 1 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

We need to solve the equations $Ly=b$ and $Ux=y$.

By Theorem 3 from Chapter 1, the matrix equation $Ly=b$ and $Ux=y$ has the same solution set as the system of equations corresponding to the augmented matrix $[L \ b]$ and $[U \ y]$.

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 2 & 1 & 0 & 0 & 7 \\ -1 & 0 & 1 & 0 & 0 \\ -4 & 3 & -5 & 1 & 3 \end{bmatrix} \xrightarrow[\text{-2} \times \text{Row 1}]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 5 \\ -1 & 0 & 1 & 0 & 0 \\ -4 & 3 & -5 & 1 & 3 \end{bmatrix} \xrightarrow[\text{+Row 1}]{\text{Row 3} = \text{Row 3}} \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 5 \\ 0 & 0 & 1 & 0 & 1 \\ -4 & 3 & -5 & 1 & 3 \end{bmatrix}$$

$$\xrightarrow[\text{+4} \times \text{Row 1}]{\text{Row 4} = \text{Row 4}} \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 5 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 3 & -5 & 1 & 7 \end{bmatrix} \xrightarrow[\text{-3} \times \text{Row 1}]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 5 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & -5 & 1 & -8 \end{bmatrix} \xrightarrow[\text{+5} \times \text{Row 3}]{\text{Row 4} = \text{Row 4}} \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 5 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -3 \end{bmatrix}$$

$$\therefore y = \begin{bmatrix} 1 \\ 5 \\ 1 \\ -3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & -4 & -3 & 1 \\ 0 & -3 & 1 & 0 & 5 \\ 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & -3 \end{bmatrix} \xrightarrow[\text{+3} \times \text{Row 4}]{\text{Row 1} = \text{Row 1}} \begin{bmatrix} 1 & -2 & -4 & 0 & -8 \\ 0 & -3 & 1 & 0 & 5 \\ 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & -3 \end{bmatrix} \xrightarrow[\text{-Row 4}]{\text{Row 3} = \text{Row 3}} \begin{bmatrix} 1 & -2 & -4 & 0 & -8 \\ 0 & -3 & 1 & 0 & 5 \\ 0 & 0 & 2 & 0 & 4 \\ 0 & 0 & 0 & 1 & -3 \end{bmatrix} \xrightarrow[\text{+4} \times \text{Row 3}]{\text{Row 1} = \text{Row 1}} \begin{bmatrix} 1 & -2 & -4 & 0 & -8 \\ 0 & -3 & 1 & 0 & 5 \\ 0 & 0 & 2 & 0 & 4 \\ 0 & 0 & 0 & 1 & -3 \end{bmatrix}$$